

ECON*6020 Macroeconomic Theory I
Assignment 2 – Fall 2026

Instructor: Stephen Kosempel
Due Date: Friday October 9, 2026

Consider an economy that consists of L identical consumers and N firms. Each consumer supplies 1 unit of labour to the firms each period. The wage rate per worker is given by w . Consumers also rent capital to firms. The rental rate of capital is given by r . The budget constraint for the representative consumer is given by

$$\dot{k} = s \times (w + rk) - \delta k,$$

where k denotes the capital-labour ratio (K/L), s is the savings rate, and δ is the depreciation rate. There is no population growth.

Each producer i creates output via a Cobb-Douglas constant returns to scale production function given by

$$Y_i = (AL_i)^{1-\alpha} K_i^\alpha,$$

where A denotes the state of technology. Arrow (1962) has endogenized the growth process in the Solow-Swan model by assuming that a producer that increases its physical capital learns simultaneously how to produce more efficiently. The positive effect of experience on productivity is called learning-by-doing, or in this case learning-by-investing. We will attempt to capture Arrow's concept of learning-by-investing by assuming that the state of technology is a function of the average capital stock in the firms,

$$A = a \left(\frac{1}{N} \sum_{i=1}^N K_i \right)^b, \text{ where } a > 0 \text{ and } b \geq 0.$$

Firms in choosing their own K_i and L_i treat A as given. Therefore, there is an externality to capital accumulation.

- (a) State the market clearing conditions in the factor markets.
- (b) Derive the competitive equilibrium real wage rate and real rental rate of capital as a function of only the model's parameters and the average capital-labour ratio in the economy.
- (c) Show that for $b > 0$ the economy as a whole has increasing returns to scale in capital and labour.
- (d) Describe the transition dynamic to the steady-state for: (i) $b < 1$, (ii) $b > 1$, and (iii) $b = 1$.

- (e) For $b=1$ what is the steady-state growth rate of the economy?
- (f) Set $\alpha=1/3$, $a=1$, $\delta=10\%$, $L/N=1$, and $s=25\%$. Derive the steady-state level of capital for $b=0$. Starting with an initial capital-labour ratio that is $1/4$ the steady-state level, simulate a series using MATLAB for the economy's growth rate of income per capita for the next 100 years. Now set $b=1/2$ and redo the experiment, but do not change the starting value. Identify the key differences in the two growth paths. Which version do you think fits the data better: $b=0$ or $b=1/2$?
- (g) For the $b=0$ case solve for the Golden Rule level of capital. Is $s=25\%$ dynamically efficient?